

Task 1: Install and run OpenVAS or Nessus

Step 1 - Update kali

- apt update

Step 2 - Install Greenbone/OpenVAS

- sudo apt install gvm -y

Then run the setup:

- sudo gvm-setup

Step 3 — Start the services

- sudo gvm-start

Step 4 — Scan your own machine

Find your Kali IP:

- ip addr

Task 2: Scan your Wi-Fi router with Nmap

On Kali:

- ip route

You might see:

- default via 192.168.182.2 dev eth0

Here:

- Router IP = 192.168.182.2

Now perform the assignment scan:

- sudo nmap -v 192.168.182.2

For a cleaner result that you can save:

- nmap -v 192.168.1.1 -oN router_scan.txt

View the result:

- cat router_scan.txt

Task 3: Identify 3 vulnerabilities/warnings

1. Missing security update — Medium: The system is running software for which a security update is available, increasing exposure to known vulnerabilities.

2. Weak TLS configuration — Medium: The service supports an outdated or weak cryptographic configuration that could reduce communication security.
3. Unnecessary network service — Low: A network service is exposed even though it may not be required, increasing the system's attack surface.

Task 4: Research a real vulnerability in a daily-use app

A good example is Zomato's 2017 security breach.

Zomato disclosed that an attacker obtained access to its internal systems and approximately 17 million user records were stolen. Zomato said the attack involved a compromised developer account and that the attacker exploited a vulnerability in older application code. Zomato subsequently reset affected passwords and investigated the incident.

How could scanning help?

A vulnerability scanner could potentially help identify:

- Outdated software/components
- Known vulnerabilities in exposed services
- Weak security configurations
- Vulnerable application components

However, a vulnerability scanner alone would not necessarily detect compromised credentials or every application-logic vulnerability. Credential monitoring, code review, penetration testing, logging, and access-control auditing may also be necessary.

Task 5: Bash script to scan three IP addresses

Create the file:

- nano scan_ips.sh

Put this inside:

```
#!/bin/bash
```

```
ips=("192.168.1.10" "192.168.1.20" "192.168.1.30")
```

```
for ip in "${ips[@]}"
```

```
do
```

```
    echo "Scanning $ip..."
```

```
    nmap -sV "$ip" -oN "scan_$ip.txt"
```

```
done
```

```
echo "All scans completed."
```

Save with:

- Ctrl + O → Enter → Ctrl + X

Make it executable:

- `chmod +x scan_ips.sh`

Run it:

- `./scan_ips.sh`